CS768 Midsem Report
Anisotropic Message Passing Analysis in Directional Graph Networks

22B0959, 22B0987, 22B1008, 22B1062

March 9, 2026

Abstract

Graph Neural Networks (GNNs) have emerged as powerful architectures for learning on graph-structured data. However, they suffer from fundamental limitations including lack of anisotropy, over-smoothing, and reduced expressiveness due to symmetric isotropic kernels. This project aims to implement Directional Graph Networks (DGNs), a framework that introduces globally consistent anisotropic kernels based on Laplacian eigenvector gradients.

We propose to conduct both implementation and qualitative analysis on small-scale benchmark datasets to understand directional message passing mechanisms. Our work will explore how vector field-based aggregation matrices enable efficient information propagation across distant nodes while mitigating over-smoothing effects. This report outlines the problem formulation, theoretical background, and implementation roadmap for studying directional aggregation operators.

1 Introduction

Convolutional Neural Networks (CNNs) leverage spatial directionality through canonical axes (horizontal and vertical), enabling the design of directional kernels such as Gabor filters that capture oriented frequency information. However, the absence of analogous directional concepts in Graph Neural Networks fundamentally limits their discriminative power. Most contemporary GNN methods rely on symmetric, isotropic aggregation kernels that treat all incoming messages equally regardless of their structural orientation within the graph.

This limitation becomes significant in domains where anisotropic structures play a crucial role. Molecular graphs exhibit directional bonding patterns, transportation networks contain flow-dependent relationships, and physical systems may have anisotropic material properties. Conventional GNN architectures cannot naturally encode these directional characteristics, leading to inefficient message propagation and limited representational capacity.

Directional Graph Networks address this limitation by introducing vector fields defined over graphs that capture globally consistent directional flows. By projecting node-level messages into these directional fields before aggregation, DGNs enable anisotropic message passing analogous to directional convolutions used in computer vision.

The use of Laplacian eigenvectors as vector fields is grounded in spectral graph theory. These eigenvectors capture essential global structural information of the graph and provide interpretable directions for information propagation. This project focuses on implementing DGNs and performing qualitative analysis of directional message passing on small benchmark datasets to understand the practical impact of directional aggregation mechanisms.

2 Problem Formulation

2.1 Core Challenge

The central challenge addressed in this work is the design of anisotropic graph convolution kernels that depend on global graph topology rather than purely local neighborhood features. Formally, given an undirected graph $G = (V, E)$ with n vertices and node feature matrix $X \in \mathbb{R}^{n\times d}$, we aim to construct aggregation operators that can:

1. Weight neighbor contributions according to their alignment with globally defined directional fields.
2. Compute both low-pass (smoothing) and high-pass (derivative) filters along specific directions.
3. Reduce diffusion distance between distant nodes to mitigate over-smoothing.
4. Generalize standard CNN kernels when applied to grid-like graphs.

2.2 Mathematical Formulation

Vector fields on graphs are defined as matrices $F \in \mathbb{R}^{n\times n}$ with the same sparsity pattern as the adjacency matrix. The entry $F_{i,j}$ represents the directional flow magnitude from node i to node j.

Two aggregation operators are used to perform directional message passing.

Directional Smoothing Matrix

$$B_{\mathrm{av}}(F)_{i,:} = \frac{|F_{i,:}|}{\|F_{i,:}\|_{L^1} + \epsilon} \tag{1}$$

Directional Derivative Matrix

$$B_{\mathrm{dx}}(F)_{i,:} = \hat{F}_{i,:} - \mathrm{diag}\left(\sum_j \hat{F}_{:,j}\right)_{i,:} \tag{2}$$

where

$$\hat{F}_{i,:} = \frac{F_{i,:}}{\|F_{i,:}\|_{L^1} + \epsilon}.$$

2.3 Vector Field Construction via Laplacian Eigenvectors

Instead of manually defining directional fields, we construct them using gradients of low-frequency Laplacian eigenvectors.

For the normalized Laplacian

$$L_{\mathrm{norm}} = D^{-1}L,$$

the eigenvector ϕ_1 corresponding to the smallest non-trivial eigenvalue λ_1 captures the dominant structural direction of the graph. The vector field can then be defined as

$$F = \nabla\phi_1.$$

This approach is theoretically grounded in spectral graph theory and has been shown to reduce diffusion distance between distant nodes in graph diffusion processes.

3 Literature Review

3.1 Graph Neural Networks and Expressiveness

Standard message-passing neural networks aggregate neighbor features using permutation-invariant operations. Graph Convolutional Networks (GCNs) perform symmetric adjacency-based aggregation, while Graph Isomorphism Networks (GINs) employ sum aggregation to maximize expressive power.

Despite these improvements, these architectures remain isotropic because they treat all neighbors equivalently. As a result, they cannot distinguish different spatial configurations of neighbors.

The Weisfeiler-Lehman (WL) graph isomorphism test provides a theoretical upper bound on the expressive power of GNNs. Many existing GNN architectures are limited to the discriminative capability of the 1-WL test.

3.2 Over-Smoothing and Over-Squashing

Deep GNN architectures suffer from the problem of over-smoothing, where node representations converge to similar values after repeated message passing steps.

Another related issue is over-squashing, where information from distant nodes becomes compressed into low-dimensional representations during propagation.

The diffusion distance between two nodes x and y at time t can be expressed as

$$d_t(x, y) = \left(\sum_{z \in V} |q_t(x, z) - q_t(y, z)|^2 \right)^{1/2}. \tag{3}$$

Using Laplacian eigenvector gradients helps reduce this diffusion distance approximately proportionally to $e^{-\lambda_1}$, improving long-range information propagation.

3.3 Directional Kernels and Spectral Foundations

Directional filters such as Gabor kernels are widely used in computer vision to capture oriented structures in images.

Graph Neural Networks lack analogous directional filters due to the irregular structure of graphs. Laplacian eigenvectors provide a principled way to construct such directions because they act as Fourier modes for graph signals.

On grid graphs, Laplacian eigenvectors correspond to cosine basis functions aligned with horizontal and vertical axes. This observation shows that Directional Graph Networks generalize CNN-style directional kernels to arbitrary graph structures.